

Re: Recommended Sample for the Ablation Zone Design

5 messages

Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>

Tue, Jul 21, 2026 at 10:52 AM

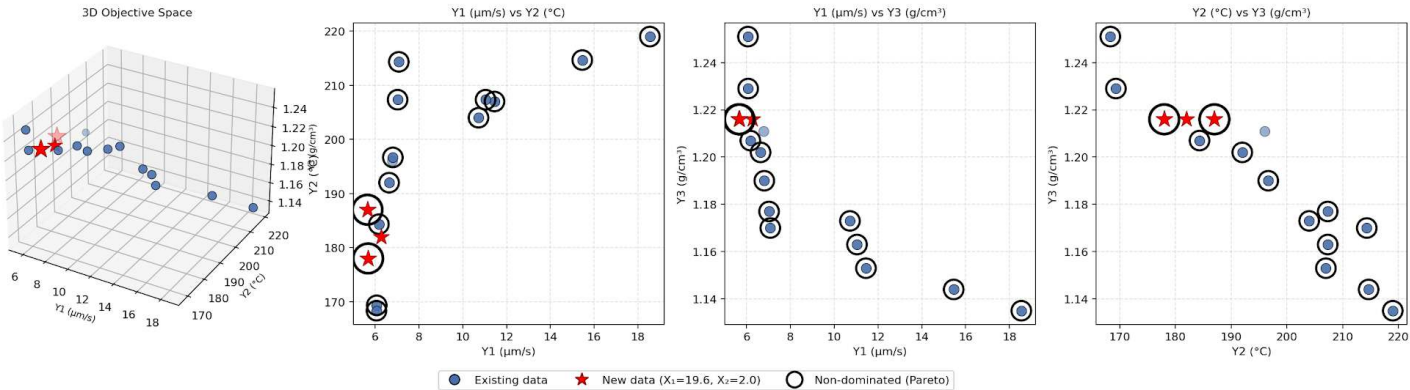
To: "Chen, Yinong" <ychen531@uic.edu>

Cc: Wei Chen <weichen@northwestern.edu>, "Pan, Yayue" <yayuepan@uic.edu>, Christian Fernandez <ChristianFernandezSoria2030@u.northwestern.edu>

Hi Yinong,

Great to see that the predicted values align well with the experimental results, though some of the Backside temperatures go out of the predicted interval. Still, the improvement in the overall objective space is noticeable. I am visualizing the objective space below, where the left is the 3-dim space and the others are projected 2-dim spaces. The three red stars represent the new samples you just sent.

Updated Pareto Front (14 solutions) | HV = 53.240 (Initial: 51.168)



- Hypervolume update: 51.168 → 53.240 (the volume covered by the samples, higher is better)
- For Y1, Replicates 1 and 2 have the lowest ablation rates observed (5.65–5.68 μm/s) so far, which is very promising.
- For Y2, the values of the new samples are not bad, two are the third and fourth lowest Backside temp.

In the meanwhile, I want to introduce Christian (copied in the email) who is also working on the MIRO project. He will take over this work after I leave. We can schedule a meeting later this week or next week. Please let me know your preferred time to meet.

Best regards,

Ying-Kuan (Rick) Tsai | Postdoctoral Scholar
Integrated Design Automation Laboratory (IDEAL)
Department of Mechanical Engineering | Northwestern University
[2145 Sheridan Road](#), Tech B224, Evanston, IL 60208

Phone: 979-969-9053 | Email: yingkuan.tsai@northwestern.edu

From: Chen, Yinong <ychen531@uic.edu>

Sent: Monday, July 20, 2026 12:17 PM

To: Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>

Cc: Wei Chen <weichen@northwestern.edu>; Pan, Yayue <yayuepan@uic.edu>

Subject: Re: Recommended Sample for the Ablation Zone Design

Hi Rick,

Thanks for the update! We've finished the ablation test today morning and here are the results:

Carbon black wt%	Fe-rich regolith wt%	I-369 wt%	Resin wt%	Linear ablative rate (μm/s)	Backside temp. (°C)	Density (g/cm³)
19.6	2	2	76.4	5.65	187	1.216
19.6	2	2	76.4	5.68	178	1.216
19.6	2	2	76.4	6.27	182	1.216

I've compared the testing results with your prediction, and it looks close. Please let me know if you have any question or if you want to arrange a meeting to discuss in more detail. Thanks!

Best,
Yinong

From: Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>
Sent: Saturday, July 18, 2026 21:13
To: Chen, Yinong <ychen531@uic.edu>
Cc: Wei Chen <weichen@northwestern.edu>; Pan, Yayue <yayuepan@uic.edu>
Subject: Re: Recommended Sample for the Ablation Zone Design

CAUTION: External Sender

Hi Yinong,

No problem. I wanted to send you the predicted mean and standard deviation of the proposed location by the GP model for your reference:

Objective	Mean (μ)	Std Dev (σ)	95% CI
Y_1 — Ablation Rate ($\mu\text{m/s}$)	5.968	1.026	[3.958, 7.979]
Y_2 — Backside Temp ($^{\circ}\text{C}$)	177.67	1.434	[174.86, 180.48]
Y_3 — Density (g/cm^3)	1.2160	0.0008	[1.2143, 1.2176]

Therefore, after you complete the experiments, you can compare these values. Feel free to let me know if you have any questions.

Best regards,

Ying-Kuan (Rick) Tsai | Postdoctoral Scholar
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Phone: 979-969-9053 | Email: yingkuan.tsai@northwestern.edu

From: Chen, Yinong <ychen531@uic.edu>
Sent: Monday, July 13, 2026 11:27 AM
To: Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>
Cc: Wei Chen <weichen@northwestern.edu>; Pan, Yayue <yayuepan@uic.edu>
Subject: Re: Recommended Sample for the Ablation Zone Design

Hi Rick,

Thanks for the update! I will work on the new sample fabrication and get back to you with the results as soon as possible.

Best,
Yinong

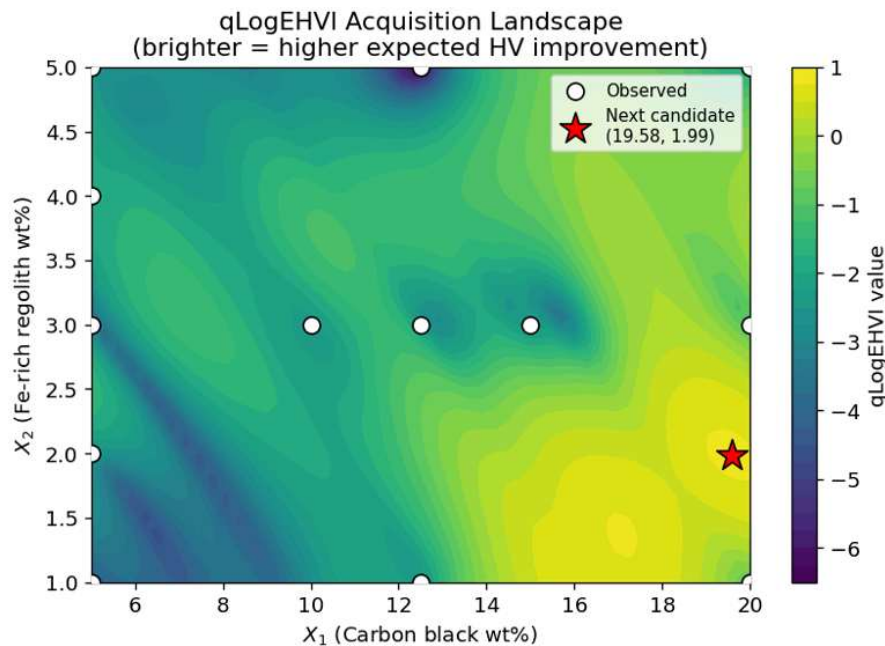
From: Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>
Sent: Friday, July 10, 2026 13:12
To: Chen, Yinong <ychen531@uic.edu>
Cc: Wei Chen <weichen@northwestern.edu>; Pan, Yayue <yayuepan@uic.edu>
Subject: Recommended Sample for the Ablation Zone Design

CAUTION: External Sender

Hi Yinong,

As we discussed last week, I am sending you the recommended sample using multi-objective Bayesian optimization (BO) with the multi-output heteroscedastic Gaussian process model. The acquisition function (i.e., criterion/objective for BO) I used is **Expected Hypervolume Improvement (EHVI)**, which is the most commonly used for multi-objective problems with noises. This higher value the EHVI is, the higher potential the sample can be.

If only one sample is selected, I will choose: $X_1 = 19.58$, $X_2 = 1.99$, whose EHVI = 0.785. Here, $X_1 \in [5, 20]$ is Carbon black (wt%), and $X_2 \in [1, 5]$ is Fe-rich regolith (wt%). The image below shows the heatmap of the EHVI value in the design space, and you see that the next candidate is located at the highest point. Since you prefer the values to the first decimal, I would say the values can be $X_1 = 19.6$, $X_2 = 2.0$.



If you would like to schedule a meeting to discuss more details, we can meet the week of 20th July since I will be gone in this coming week. On the other hand, feel free to email me if you have any questions.

Best regards,

Ying-Kuan (Rick) Tsai | Postdoctoral Scholar
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Chen, Yinong <yichen531@uic.edu>

Tue, Jul 21, 2026 at 11:11 AM

To: "Ying-Kuan (Rick) Tsai" <yingkuan.tsai@northwestern.edu>

Cc: Wei Chen <weichen@northwestern.edu>, "Pan, Yayue" <yayuepan@uic.edu>, Christian Fernandez <ChristianFernandezSoria2030@u.northwestern.edu>

Hi Rick,

Thank you so much for this promising update! I'm available this Wednesday and next Tuesday to Friday, 9am to 5pm. Please let me know what time works for you and Christian.

Best,
Yinong

From: Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>

Sent: Tuesday, July 21, 2026 10:52

To: Chen, Yinong <yichen531@uic.edu>

Cc: Wei Chen <weichen@northwestern.edu>; Pan, Yayue <yayuepan@uic.edu>; Christian Fernandez <ChristianFernandezSoria2030@u.northwestern.edu>

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[Quoted text hidden]

Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>

Tue, Jul 21, 2026 at 12:49 PM

To: "Chen, Yinong" <yichen531@uic.edu>

Cc: Wei Chen <weichen@northwestern.edu>, "Pan, Yayue" <yayuepan@uic.edu>, Christian Fernandez <ChristianFernandezSoria2030@u.northwestern.edu>

Hi Yinong,

Let's meet at 2 PM next Tuesday. Both Chrisitan and I will join the meeting. I will send you the calendar invite with the zoom link.

Best regards,

Ying-Kuan (Rick) Tsai | Postdoctoral Scholar
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Phone: 979-969-9053 | Email: yingkuan.tsai@northwestern.edu

From: Chen, Yinong <ychen531@uic.edu>
Sent: Tuesday, July 21, 2026 11:11 AM
To: Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>

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[Quoted text hidden]

Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>
To: "Chen, Yinong" <ychen531@uic.edu>
Cc: Christian Fernandez <ChristianFernandezSoria2030@u.northwestern.edu>, Doksoo Lee <doksoo.lee@northwestern.edu>

Tue, Jul 28, 2026 at 2:09 PM

Hi Yinong,

Are you going to join the meeting?
<https://northwestern.zoom.us/j/96582287760>

Best regards,

Ying-Kuan (Rick) Tsai | Postdoctoral Scholar
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Department of Mechanical Engineering | Northwestern University
[2145 Sheridan Road](#), Tech B224, Evanston, IL 60208

Phone: 979-969-9053 | Email: yingkuan.tsai@northwestern.edu

From: Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>
Sent: Tuesday, July 21, 2026 12:49 PM

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Chen, Yinong <ychen531@uic.edu>
To: "Ying-Kuan (Rick) Tsai" <yingkuan.tsai@northwestern.edu>
Cc: Christian Fernandez <ChristianFernandezSoria2030@u.northwestern.edu>, Doksoo Lee <doksoo.lee@northwestern.edu>

Tue, Jul 28, 2026 at 2:12 PM

Hi Rick,

So Sorry about that! I'm stuck in the airport without my laptop... I am trying to connect to the meeting using my phone... give me 2 more min to find a good network connection...

获取 [Outlook for iOS](#)

发件人: Ying-Kuan (Rick) Tsai <yingkuan.tsai@northwestern.edu>
发送时间: Tuesday, 28 July 2026 14:09:25
收件人: Chen, Yinong <ychen531@uic.edu>
抄送: Christian Fernandez <ChristianFernandezSoria2030@u.northwestern.edu>; Doksoo Lee <doksoo.lee@northwestern.edu>
主题: Re: Recommended Sample for the Ablation Zone Design

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